Silkali Sheet Cutting Optimiser — Updated Project Summary

Project Overview

The Silkali Sheet Cutting Optimiser is a browser-based warehouse planning tool designed to help optimise the cutting of 8ft x 4ft boards into smaller requested pieces while accounting for:

- Saw blade thickness (kerf)
- Additional spacing between cuts
- Material efficiency
- Manual operator adjustments
- Board offcuts
- Exportable layouts
- Saved reusable plans

The tool was intentionally built as a lightweight single-file application using HTML, CSS and JavaScript so it can:

- Run locally without installation
- Be hosted freely via GitHub Pages
- Be opened from SharePoint/Teams links
- Be maintained internally without needing infrastructure
- Work entirely client-side without a backend

Original Goals

The initial purpose of the project was to:

1. Reduce wasted board material
2. Standardise warehouse cutting layouts
3. Visualise board usage before cutting
4. Account for real-world saw blade thickness
5. Provide an easy-to-use warehouse-facing interface
6. Avoid requiring specialist software
7. Allow quick experimentation with different layouts
8. Enable warehouse feedback-driven iteration

The project later expanded into:

- Manual drag-and-drop layout editing
- Persistent saved plans
- Snap alignment behaviour
- Manual board rearrangement
- Adjustable spacing controls
- Exportable layout images
- Interactive operator workflow

Current Application Features

1. Preset Size Library

The system contains a large preset catalogue of common sheet sizes.

Each preset includes:

- Imperial dimensions (feet)
- Matching metric reference (mm)

Examples:

- 1 x 1 → 305 x 305mm
- 6 x 2 → 1830 x 610mm
- 8 x 4 → 2440 x 1220mm
- 2000 x 500 custom-format entries

The dropdown automatically renders both formats for operator clarity.

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2. Cut Queue / Priority System

Operators can:

- Add cuts to a queue
- Define quantity
- Enable/disable rotation permission
- Remove lines
- Increase/decrease quantities inline
- Edit quantities manually

Additional logic:

- Duplicate cuts merge automatically
- Prevents accidental repeated entries
- Simplifies operator workflow

Example:

Adding:

- 4x2 Qty 2
- 4x2 Qty 3

Automatically becomes:

- 4x2 Qty 5

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3. Optimisation Engine

The optimiser performs automatic layout generation.

Current logic:

- Boards initialise as full stock rectangles
- Cuts are sorted by descending area
- Best-fit placement is selected
- Remaining board space becomes reusable stock regions
- Kerf and spacing are applied to generated split regions

This uses a rectangle-splitting methodology.

Key behaviour:

- Larger pieces placed first
- Attempts to minimise waste
- Supports rotation when enabled
- Supports multiple boards

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4. True Kerf Model

A major warehouse-driven improvement.

The optimiser now models:

- Saw blade thickness (kerf)
- Real cutting loss
- Separation between pieces

The warehouse explained that cuts are not made at nominal size.

Example:

A 1ft cut is not cut at exactly 305mm.

Instead:

- Blade thickness is divided across cut boundaries
- Prevents cumulative loss across repeated cuts
- Prevents final pieces becoming undersized

The application models this by:

- Expanding spacing between generated cut regions
- Applying kerf during stock splitting
- Maintaining valid cutting clearance during manual dragging

Current default:

- 6mm kerf

Fully adjustable in the UI.

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5. Additional Gap Control

Operators can define extra spacing beyond kerf.

Purpose:

- Safety margins
- Machine clearance
- Material handling allowance
- Protective spacing

The optimiser treats:

Effective spacing = kerf + extra gap

This affects:

- Automatic placement

- Collision detection
- Snap alignment
- Manual movement validation

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6. Snap System

The tool includes adjustable snapping.

Operators define:

- Snap distance in mm

The system then:

- Snaps pieces to edges
- Snaps pieces to neighbouring cuts
- Preserves kerf/gap spacing
- Helps maintain clean alignment

This significantly improved usability during manual rearrangement.

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7. Manual Drag-and-Drop Editing

One of the largest upgrades.

Operators can:

- Drag pieces freely
- Move pieces between boards
- Rearrange layouts manually
- Fine-tune optimiser results

The system validates:

- Board boundaries
- Piece overlap
- Kerf spacing
- Gap spacing

Invalid placements:

- Highlight in red

- Revert automatically on release

This allows human override while preserving valid manufacturing constraints.

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8. Piece Rotation

Double-clicking a piece rotates it.

Rotation only works when:

- Rotation was enabled for that cut line

Rotation logic:

- Swaps width/height
- Re-validates spacing
- Re-validates collisions
- Prevents illegal placements

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9. Multi-Board Support

The optimiser supports:

- Multiple full boards
- Automatic overflow handling
- Board-specific placement tracking

Each board is rendered independently.

Visual labels:

- Board 1
- Board 2
- Board 3
- etc.

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10. Offcut Tracking

The optimiser tracks unused stock regions.

Displayed as:

- Remaining offcut rectangles
- Width x height values

Important limitation:

Current offcuts only reflect:

- The last optimiser run

Manual dragging does not currently recompute offcuts.

This was intentionally accepted as a trade-off to preserve stability.

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11. Material Efficiency Calculation

The tool calculates:

Material efficiency (%)

Formula:

Used cut area / total board area

Purpose:

- Compare layouts
- Measure waste reduction
- Benchmark optimisation quality

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12. Recommended Cutting Order

The application generates:

Recommended cutting order

Current strategy:

- Largest area first

Purpose:

- Reduce fragmentation
- Improve packing success
- Guide warehouse workflow

13. Export Layout Image

The application can export:

- Full board layouts
- As JPG images

Implemented via:

- html2canvas

Used for:

- Warehouse sharing
- Printouts
- Teams messages
- Documentation
- Cutting instructions

14. Layout Freshness Protection

A hidden stability feature.

The application detects when:

- Cuts changed
- Kerf changed
- Gap changed
- Board count changed

If changes occur:

- Layout becomes marked as outdated
- Export is blocked until re-optimised

Purpose:

Prevent operators exporting stale layouts accidentally.

15. Persistent Save System

The latest version includes:

- Named save slots
- Multiple saved plans
- Load/save/delete workflow
- Overwrite support

Stored using:

- Browser localStorage

Benefits:

- No backend required
- Instant save/load
- Multiple reusable jobs
- Warehouse-friendly workflow

Each save stores:

- Cuts
- Placements
- Stock
- Board settings
- Kerf
- Gap
- Snap settings
- Layout freshness state

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16. Save Roster Interface

The save system includes:

- Named save dropdown
- Save timestamps
- Newest-first ordering
- Status feedback messages

Operators can:

- Create named plans

- Reload previous jobs
- Overwrite updated layouts
- Delete obsolete plans

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17. Visual Interaction Improvements

Several UI improvements were added:

- Cleaner panel layout
- Board grouping
- Improved spacing
- Better readability
- Visual drag feedback
- Invalid placement highlighting
- Responsive board container
- Button grouping
- Operator guidance text

Important Technical Behaviours

Rectangle Splitting Logic

When a piece is placed:

The remaining board area is split into:

- Right-side rectangle
- Bottom rectangle

This becomes reusable stock.

This is the core optimisation model.

Collision Detection

During dragging and optimisation:

The system checks:

- Board boundaries
- Piece overlap
- Kerf spacing
- Gap spacing

Invalid placements are rejected.

Smart Edge Snapping

The system can snap to:

- Board edges
- Adjacent piece edges
- Kerf-aware spacing offsets

This improves manual precision significantly.

Known Constraints / Accepted Trade-Offs

1. Offcuts Not Recomputed After Manual Dragging

Accepted intentionally.

Reason:

Dynamic recomputation became unstable and overly complex.

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2. Optimiser Is Heuristic-Based

The optimiser uses:

- Best-fit heuristics

Not:

- Mathematical perfect nesting
- AI packing
- Industrial CNC optimisation

Meaning:

Some layouts may still benefit from human rearrangement.

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3. Browser-Based Persistence

Current saves use:

- localStorage

Meaning:

- Saves are browser/device-specific
- Clearing browser storage removes saves

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4. Manual Layout Editing Exists Outside Optimiser

Once manually adjusted:

- The optimiser does not re-solve around those edits

This is currently acceptable for warehouse usage.

Features Attempted But Rejected

Several experimental ideas were tested and intentionally removed.

Rejected Features

Grey offcut overlays

Rejected because:

- Irregular geometry produced misleading visuals
- Difficult to interpret practically

Operator confidence score

Rejected because:

- Did not provide operational value
- Added complexity without helping decisions

Visual cut path overlays

Rejected because:

- Became visually noisy
- Hard to maintain accurately
- Did not improve warehouse usability

Forced optimiser locking

Rejected because:

- Warehouse operators need freedom to edit layouts
- Re-optimisation workflows must remain flexible

Hosting Strategy

The project is intentionally deployable via:

- GitHub Pages
- SharePoint links
- OneDrive-hosted access
- Teams tabs
- Local browser execution

GitHub Pages became the preferred hosting option because:

- Free
- Simple
- Publicly accessible
- Easy updates
- No infrastructure
- No installation required

Current Project Status

The current application successfully delivers:

- Real warehouse usability
- Kerf-aware optimisation

- Manual layout control
- Save/load workflows
- Exportable layouts
- Adjustable spacing rules
- Interactive board editing
- Browser-only deployment

The project evolved from a simple optimiser into a lightweight interactive warehouse cutting planning system.

Recommended Next Phase

Conceptually, the next potential goals are:

- Canvas-based interaction engine
- Real-time offcut recomputation
- Dynamic cut-path generation
- Undo/redo history
- Multi-user layouts
- Touchscreen support
- CNC export support (if and for, whatever format the saw allows for)
- Print-ready cutting sheets
- Barcode/job references
- Operator annotations
- Live stock inventory integration

However, the project is now approaching the limits of plain HTML/JavaScript. It requires practical application and testing and the next steps adapted with that feedback in mind.

Overall Outcome

The project successfully transformed warehouse feedback into a functioning optimisation platform that balances:

- Automation
- Real-world cutting behaviour
- Operator flexibility
- Ease of deployment

- Low operational overhead

while remaining:

- Fully browser-based
- Easy to distribute
- Easy to update
- Free to host
- Accessible without IT involvement.